

Lab Assignment 2

[08/09/2026]

[Output Statements, Assignment Statements, Data Types]

- Q.1)** Write a program to read your name from the keyboard and display it on the screen with a welcome message.
- Q.2)** Write a program to read an integer from the keyboard and display it on the screen.
- Q.3)** Write a program in Python for data type conversion from:
- int to string
 - int to float
 - string to int
 - string to float
 - float to int
 - float to string
- Q.4)** Write a program in Python to round off the value of a number.
- Q.5)** Write a program in Python to find the value of a number with two digits after decimal.
- Q.6)** Write a program in Python to left, right and centered justify a string.
- Q.7)** Write a program in Python to take a number from user and find its square root and display its binary, hex and octal equivalent.
- Q.8)** Write a program in Python to print a character 5 times.
- Q.9)** Write a program in Python to find the character values for different numbers and ASCII value for different characters.
- Q.10)** Write a program in Python to enter a sentence and calculate its length including whitespaces and excluding whitespaces. Also count number of words in that sentence.
- Q.11)** Write a program in Python to evaluate the expression of string. '100*5+9+5**2'
- Q.12)** Write a program in Python to generate a random number.
- Q.13)** Write a program in Python to randomly generate a date of month (random number generation within range (1,31)).
- Q.14)** Write a program in Python to print the current date.
- Q.15)** Write a program to evaluate the expression: $5x^2 + 10x + 5$ and print the result. Take $x=10$, $x=20$, $x=15$
- Q.16)** Write a program in Python which inputs time in hours and minutes and converts it into seconds.

- Q.17)** Write a program in Python that inputs data (class, age, marks in mathematics, science) about two persons- Ram and Shyam. The program uses relational operators to check whether they study in same class? Is Ram younger to Shyam? Is Ram's marks in mathematics more than Shyam? Is Ram's marks in science less than Ram? The answer by the program is either True or False. The operators to be used are: <, >, ==.
- Q.18)** Write a program to read a positive integer X and verify whether:
- (1) X is divisible by 2.
 - (2) X is divisible by 3.
 - (3) X is divisible by both 2 and 3.
 - (4) X is divisible 2 or 3.
- Q.19)** Write a program in Python that takes side of a square (floating-point value) as input and prints the area and perimeter of the square.
- Q.20)** Write a program for the following problem. A transportation company running a bus between two stations S and T. On the route from S to T the bus stops at intermediate stations A, B, C, D, and E in the same order. On a particular day, they have observed the following pattern in passengers traveling through the bus from S to T. Let X (>10) be the number of passengers initially in the bus at source S.
- At stage A, 10 passengers got down from the bus, and 12 passengers boarded into the bus.
 - At stage B, 15 are boarded into the bus after 10 got down from it.
 - At stage C, 5 are boarded into the bus after 2 getting off from it.
 - At stage D, 22 passengers got down from the bus, and 28 passengers boarded into the bus.
 - Finally, at stage E, 15 are boarded into the bus after 7 getting off from it."

Compute the number of passengers traveled between every two stations on the route for a given X > 10. (Read X as an input from the user.)